

1. Write a shell command that checks whether the file /etc/passwd exists and prints "yes" or "no".

```
GNU nano 0.2 1.sh
#!/bin/bash

if [ -e /etc/passwd ]; then
echo "Yes"
else
echo "no"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 1.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 1.sh
Yes
```

2. Create a script that checks whether a given directory is writable and prints an appropriate message.

```
#!/bin/bash

DIR_PATH="/etc/passwd"

if [ -w "$DIR_PATH" ]; then
echo "writable"
else
echo "not writable"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 2.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 2.sh
not writable
```

3. Using [], test whether a variable contains an empty string and print an appropriate message.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ [ -z "$v1" ] && echo "Empty string" || echo "Non Empty string"
Empty string
```

4. Write an if statement that checks whether the current user is root.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ [ "$whoami" = "root" ] && echo "root user" || echo "not a root user"
not a root user
```

5. Write a script that reads an integer and prints whether it is positive, negative, or zero.

```
#!/bin/bash

read -p "enter a number: " num

if [ $num -lt 0 ]; then
echo "number is negative"
elif [ $num -gt 0 ]; then
echo "number is positive"
else
echo "number is zero"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 5.sh
enter a number: 5
number is positive
```

6. Create a script that accepts two strings as arguments and checks whether they are equal.

```
#!/bin/bash

str1="$1"
str2="$2"

if [ "$str1" = "$str2" ]; then
echo "strings are equal"
else
echo "strings are not equal"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 6.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 6.sh "abc" "abc"
strings are equal
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

7. Use test or [] to check whether /tmp is a directory.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ [ -d "/tmp" ] && echo "its a directory" || echo "not a directory"
its a directory
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

8. Write a script that checks whether a given filename is a regular file, directory, or does not exist.

```
#!/bin/bash

file="$1"

if [ -f "$file" ]; then
echo "regular file"
elif [ -d "$file" ]; then
echo "directory"
elif [ -e "$file" ]; then
echo "file exists"
else
echo "file does not exists"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 8.sh 1.sh
regular file
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 8.sh /tmp
directory
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

9. Write a for loop that prints the names of all .txt files in the current directory.

```
#!/bin/bash

for file in *.txt;
do
echo "$file"
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 9.sh
test.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

10. Write a for loop that prints the numbers from 1 to 20.

```
#!/bin/bash

for i in {1..20}
do
echo "$i"
done
```

11. Write a while loop that prints the numbers from 1 to 10.

```
#!/bin/bash

i=1
while [ $i -le 10 ]
do
echo "$i"
((i++))
done
```

12. Write an until loop that keeps asking the user for a password until the correct password is entered.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 12.sh
enter the passworddga
enter the right password
enter the passwordsdafa
enter the right password
enter the passwordsecret123
enter the right password
access granted
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

13. Write a script using a for loop to calculate the sum of numbers from 1 to N.

```
#!/bin/bash

n=$1
sum=0
for ((i=1;i<=n;i+=1));
do
sum=$((sum+i))
done

echo "sum is $sum"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 13.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ./13.sh 4
sum is 10
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

14. Write a script to calculate the factorial of a given number using a loop.

```
#!/bin/bash

num=$1
fact=1
for((i=1;i<=$num;i++));
do
fact=$((fact*i))
done

echo "factorial is $fact"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ./14.sh 2
factorial is 2
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ./14.sh 5
factorial is 120
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 14.sh
```

15. Write a script that counts the number of files in a given directory using a loop.

```
#!/bin/bash
path="$1"
if [ ! -d "$path" ]; then
echo "Not a valid directory"
exit 1
fi
count=0
for file in "$path"/*
do
if [ -f "$file" ]; then
count=$((count+1))
fi
done
echo "number of files in the directory is $count"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 15.sh ../Documents
number of files in the directory is 26
```

16. Write a script that prints all even numbers between 1 and 100 using a loop.

```
#!/bin/bash
for i in {1..100}
do
if [ $((i%2)) = 0 ]
then
echo "$i"
fi
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 16.sh
2
4
6
8
10
12
14
```

17. Write a script that checks each number from 1 to N and prints whether it is even or odd.

```
#!/bin/bash
read -p "Enter the value of N: " n
for((i=1;i<=n;i++))
do
if [ $((i%2)) = 0 ]
then
echo "$i is even"
else
echo "$i is odd number"
fi
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 17.sh
Enter the value of N: 5
1 is odd number
2 is even
3 is odd number
4 is even
5 is odd number
```

18. Write a script that accepts a filename and counts how many lines in the file are non-empty.

```
#!/bin/bash

file_name="$1"

grep -c '^[[:space:]]' "$file_name"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 18.sh 17.sh
11
```

19. Write a shell command to start sleep 100 in the background and display the running job.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 100 & jobs
[1] 5436
[1]+  Running                  sleep 100 &
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

20. Start a long-running command in the background and demonstrate how to use jobs to display it.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ while true; do echo "Task running at $(date)" >> process.log; sleep 2; done &
[2] 5479
[1] Done                  sleep 100
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ jobs
[2]+  Running              while true; do
    echo "Task running at $(date)" >> process.log; sleep 2;
done &
```

21. Start a background job and use fg to bring it to the foreground.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 15 &
[3] 5611
[2] Terminated          while true; do
    echo "Task running at $(date)" >> process.log; sleep 2;
done
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ fg %1
bash: fg: %1: no such job
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ fg %3
sleep 15
```

22. Stop a foreground job using Ctrl+Z, then use bg to continue it in the background.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 30
^Z
[1]+  Stopped                  sleep 30
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ bg %1
[1]+ sleep 30 &
```

23. Start two background sleep processes and use jobs to identify both jobs.

```
[1]+  Stopped                  sleep 30
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 50 &
[1] 5694
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 40 &
[2] 5696
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ jobs
[1]-  Running                  sleep 50 &
[2]+  Running                  sleep 40 &
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

24. Start a background process, find its PID using ps, and terminate it using kill.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 100 &
[1] 5739
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps
  PID TTY          TIME CMD
  4379 pts/0        00:00:00 bash
  5739 pts/0        00:00:00 sleep
  5740 pts/0        00:00:00 ps
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ kill 5739
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ jobs
[1]+  Terminated              sleep 100
```

25. Start a process and send it SIGTERM using kill.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 50 &
[1] 5832
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps
  PID TTY          TIME CMD
  4379 pts/0        00:00:00 bash
  5832 pts/0        00:00:00 sleep
  5833 pts/0        00:00:00 ps
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ kill -SIGTERM 5832
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps
  PID TTY          TIME CMD
  4379 pts/0        00:00:00 bash
  5835 pts/0        00:00:00 ps
[1]+  Terminated              sleep 50
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

26. Start a process and demonstrate the difference between kill PID and kill -9 PID.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 100 &
[1] 5875
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 90 &
[2] 5876
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps
  PID TTY          TIME CMD
 4379 pts/0    00:00:00 bash
 5875 pts/0    00:00:00 sleep
 5876 pts/0    00:00:00 sleep
 5878 pts/0    00:00:00 ps
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ kill 5875
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ kill -9 5876
[1]-  Terminated                  sleep 100
```

27. Use ps to display all processes belonging to the current user.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps -u $(whoami)
  PID TTY          TIME CMD
 2258 ?            00:00:00 systemd
 2259 ?            00:00:00 (sd-pam)
 2265 ?            00:00:00 pipewire
 2266 ?            00:00:00 pipewire-media-
```

28. Use ps to display PID, PPID, CPU usage, memory usage, and command name.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps aux
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.0  0.0 168384 13848 ?        Ss   02:51   0:02 /sbin/init splash
root             2  0.0  0.0      0     0 ?        S    02:51   0:00 [kthreadd]
root             3  0.0  0.0      0     0 ?        S    02:51   0:00 [pool_workqueue_release]
root             4  0.0  0.0      0     0 ?        I<   02:51   0:00 [kworker/R-rcu_g]
root             5  0.0  0.0      0     0 ?        I<   02:51   0:00 [kworker/R-rcu_p]
root             6  0.0  0.0      0     0 ?        I<   02:51   0:00 [kworker/R-slub_]
root             7  0.0  0.0      0     0 ?        I<   02:51   0:00 [kworker/R-netns]
root             9  0.0  0.0      0     0 ?        I<   02:51   0:00 [kworker/0:0H-events_highpri]
root            12  0.0  0.0      0     0 ?        I<   02:51   0:00 [kworker/R-mm_pe]
root            13  0.0  0.0      0     0 ?        I    02:51   0:00 [rcu_tasks_kthread]
root            14  0.0  0.0      0     0 ?        I    02:51   0:00 [rcu_tasks_rude_kthread]
root            15  0.0  0.0      0     0 ?        I    02:51   0:00 [rcu_tasks_trace_kthread]
root            16  0.0  0.0      0     0 ?        S    02:51   0:00 [ksoftirqd/0]
```

29. Use ps and grep to find whether a particular process is running.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 50 &
[1] 6023
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps aux | grep 6023
pgcp-esd      6023  0.0  0.0  8372  2144 pts/0    S    04:06   0:00 sleep 50
pgcp-esd      6026  0.0  0.0  9080  2672 pts/0    S+   04:06   0:00 grep --color=auto 6023
```

30. Write a command to count how many processes are currently running on the system.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps aux | wc -l
360
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

31. Use ps and sort to display the top 5 processes by CPU usage.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps ao pcpu,pid,user --sort=-pcpu | head -n 6
%CPU    PID  USER
0.0      2322 pgcp-esd
0.0      2325 pgcp-esd
0.0      4379 pgcp-esd
0.0      6188 pgcp-esd
0.0      6189 pgcp-esd
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

32. Use ps and sort to display the top 5 processes by memory usage.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps ao pmem,pid,user --sort=-pmem | head -n 6
%MEM     PID  USER
0.0       2325 pgcp-esd
0.0       4379 pgcp-esd
0.0       2322 pgcp-esd
0.0       6262 pgcp-esd
0.0       6263 pgcp-esd
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

33. Use ps and grep to find all processes whose command contains the word bash.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ps aux | grep 'bash'
pgcp-esd  4379  0.0  0.0 12052  6448 pts/0    Ss   02:57   0:00 bash
pgcp-esd  6289  0.0  0.0  9080  2672 pts/0    S+   04:20   0:00 grep --color=auto bash
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

34. Write a script that checks every 5 seconds whether a specified process is running.

```
#!/bin/bash

pname="$1"

while true; do
if ps | grep "$pname" | grep -v "grep" > /dev/null; then
echo "$pname is running"
else
echo "$pname is not running"
fi
sleep 5
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 100 &
[1] 4826
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 34.sh 4826
4826 is running
4826 is running
4826 is running
4826 is running
```

35. Write a script that asks the user for a process name and displays matching processes using ps and grep.

```
#!/bin/bash

pname="$1"

if [ -z "$pname" ]; then
    exit 1
fi

ps aux | grep -i "$pname" | grep -v "grep"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sleep 50 &
[1] 5730
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 35.sh 5730
pgcp-esd 5730 0.0 0.0 8372 2140 pts/0 S 21:24 0:00 sleep 50
pgcp-esd 5732 0.0 0.0 9972 3632 pts/0 S+ 21:24 0:00 /bin/bash 35.sh 5730
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

36. Write a script that accepts a filename and prints whether the file is readable, writable, and executable.

```
#!/bin/bash

read -p "enter a valid file name or path to file : " file

if [ ! -f "$file" ]; then
    echo "not a valid file"
    exit 1
fi

if [ -r "$file" ]; then
    echo "file is readable"
fi
if [ -w "$file" ]; then
    echo "file is writable"
fi
if [ -e "$file" ]; then
    echo "file is executable"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 36.sh
enter a valid file name or path to file : 35.sh
file is readable
file is writable
file is executable
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

37. Create a script that accepts multiple filenames as arguments and reports whether each file exists.

```
#!/bin/bash

read -p "enter the number of files : " num

for((i=1;i<=num;i+=1))
do
    read -p "enter the name of the file : " fname
    if [ -f "$fname" ]; then
        echo "File exists"
    else
        echo "file does not exist"
    fi
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 37.sh
enter the number of files : 3
enter the name of the file : 3.txt
file does not exist
enter the name of the file : test.txt
File exists
enter the name of the file : 37.sh
File exists
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

38. Write a script that accepts a directory name as an argument and displays all files larger than 1 MB.

```
GNU nano 6.2 38.sh
#!/bin/bash

dir="$1"

for file in "$dir"/*;
do
if [ -f "$file" ] && [ "$(wc -c < "$file")" -gt 1000000 ]; then
ls -l "$file"
fi
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 38.sh ../Downloads
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1150447 Sep 10 18:52 ../Downloads/5018_linux-Assignment_3.pdf
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1690606 Sep 9 19:34 '../Downloads/5018_Linux_Assignment_SLA_1(1).odt'
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1690606 Sep 9 17:56 ../Downloads/5018_Linux_Assignment_SLA_1.odt
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1241020 Sep 9 17:58 ../Downloads/5018_Linux_Assignment_SLA_1.pdf
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1544145 Sep 1 22:02 ../Downloads/5019_Linux_assignment_1.pdf
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1150597 Sep 10 18:48 ../Downloads/5019_linux-Assignment_3.pdf
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1289738 Sep 9 19:37 ../Downloads/5019_Linux_Assignment_SLA_1-1.pdf
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1289738 Sep 9 19:35 ../Downloads/5019_Linux_Assignment_SLA_1.pdf
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1114927 Sep 13 04:28 ../Downloads/Linux_assignment-4.odt
-rw-rw-r-- 1 pgcp-esd pgcp-esd 11900117 Sep 10 21:52 ../Downloads/samplerestaurant.json
-rw-rw-r-- 1 pgcp-esd pgcp-esd 1762525 Sep 11 16:18 ../Downloads/thumb-1920-1240656.png
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

39. Write a script that accepts a directory name and counts the number of directories inside it.

```
#!/bin/bash

dir="$1"
count=0

for file in "$dir"/*;
do
if [ -d "$file" ]; then
count=$((count+1))
fi
done

echo "there are $count directories inside $dir"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 39.sh ../Documents
there are 15 directories inside ../Documents
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

40. Write a script that reads a file line by line and prints each line with its line number.

```
#!/bin/bash

file="$1"
Line_num=1

while IFS= read -r line;
do
echo "$Line_num: $line"
((Line_num++))
done < "$file"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 40.sh 39.sh
1: #!/bin/bash
2:
3: dir="$1"
4: count=0
5:
6: for file in "$dir"/*;
7: do
8: if [ -d "$file" ]; then
9: count=$((count+1))
10: fi
11: done
12:
13: echo "there are $count directories inside $dir"
14:
```

41. Write a script that reads a file and counts the number of lines containing the word error using grep.

```
#!/bin/bash

file="$1"

grep -ic "error" "$file"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano test.txt
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 41.sh test.txt
3
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

42. Write a command using grep to search for a word in a text file, ignoring case.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 39.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ grep -i file 39.sh
for file in "$dir"/*;
if [ -d "$file" ]; then
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

43. Use grep to display only those lines from /etc/passwd that contain the word bash.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ grep -i bash /etc/passwd
root:x:0:0:root:/root:/bin/bash
pgcp-esd:x:1000:1000:PGCP-BDA,,,:/home/pgcp-esd:/bin/bash
exam:x:1001:1001:exam,,,:/home/exam:/bin/bash
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

44. Use grep to search recursively for the word password in all files under a given directory.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ grep -r password ../Documents
../Documents/exercise2.sh:read -sp "enter the password: " password
../Documents/exercise2.sh:if [[ ( "$username" == "admin" && "$password" == "secret123" ) || ( "$username" == "root" && "$password" == "rootpass" ) ]]; then echo "login Successfull"
grep: ../Documents/directory_by_dev1: Permission denied
grep: ../Documents/permission_test/a.sh: Permission denied
grep: ../Documents/permission_test/b.sh: Permission denied
grep: ../Documents/developers_directory: Permission denied
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

45. Use grep with line numbers to display matching lines from a file.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ grep -n "file" 40.sh
3:file="$1"
10:done < "$file"
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

46. Use grep to count how many lines in a file contain a given word.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ grep -c "file" 40.sh
2
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

47. Write a command using sort to sort the contents of a text file alphabetically.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sort 39.sh

#!/bin/bash
count=$((count+1))
count=0
dir="$1"
do
done
echo "there are $count directories inside $dir"
fi
for file in "$dir"/*;
if [ -d "$file" ]; then
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

48. Use sort -n to sort a file containing numbers in ascending order.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sort -n test.txt
14
56
423
642
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

49. Use sort -r to sort a list in descending order.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sort -r 41.sh
grep -ic "error" "$file"
file="$1"
#!/bin/bash

pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

50. Use sort together with uniq to display unique values from a file.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sort -n -u test.txt
14
56
423
564
642
```

51. Write a command to sort /etc/passwd based on the username field.

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ sort -t ":" -k 1,1 /etc/passwd
analyst1:x:1006:1006::/home/analyst1:/bin/sh
_apt:x:105:65534::/nonexistent:/usr/sbin/nologin
avahi:x:114:121:Avahi mDNS daemon,,,:/run/avahi-daemon:/usr/sbin/nologin
avahi-autoipd:x:110:119:Avahi autoip daemon,,,:/var/lib/avahi-autoipd:/usr/sbin/nologin
backup:x:34:34:backup:/var/backups:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
cassandra:x:129:137:Cassandra database,,,:/var/lib/cassandra:/usr/sbin/nologin
colord:x:123:130:colord colour management daemon,,,:/var/lib/colord:/usr/sbin/nologin
cups-pk-helper:x:115:122:user for cups-pk-helper service,,,:/home/cups-pk-helper:/usr/sbin/nologin
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
```

52. Create a script that accepts three numbers and prints the largest number using if-elif-else.

```
#!/bin/bash

read -p "Enter the three numbers : " n1 n2 n3

if [ $n1 -gt $n2 ] && [ $n1 -gt $n3 ]; then
echo "$n1 is the largest"
elif [ $n2 -gt $n1 ] && [ $n2 -gt $n3 ]; then
echo "$n2 is the largest"
else
echo "$n3 is the largest"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 52.sh
Enter the three numbers : 4 5 6
6 is the largest
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

53. Write a script using case that accepts start, stop, and status and prints a corresponding message.

```
#!/bin/bash

read -p "Enter start/stop/status : " comm

case "$comm" in
"start") echo "you selected start";;
"stop") echo "you selected stop";;
"status") echo "you selected status";;
*) echo "invalid input";;
esac
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 53.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 53.sh
Enter start/stop/status : stop
you selected stop
```

54. Create a simple menu-driven script using case for options such as Create File, Delete File, Display File, and Exit.

```
#!/bin/bash

while true;
do

read -p "Enter 'Create'/'Delete'/'Display'/'Exit' : " comm
case "$comm" in
"Create") read -p "enter the name of the file " name
touch "$name";;
"Delete") read -p "enter the name to delete " del
rm "$del";;
"Display") read -p "enter the name to display" dis
cat "$dis";;
"Exit") exit 1;;
*) echo "invalid input";;
esac
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 54.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 54.sh
Enter 'Create'/'Delete'/'Display'/'Exit' : Create
enter the name of the file abc.txt
Enter 'Create'/'Delete'/'Display'/'Exit' : Delete
enter the name to delete abc.txt
Enter 'Create'/'Delete'/'Display'/'Exit' : Display
enter the name to display abc.txt
cat: abc.txt: No such file or directory
Enter 'Create'/'Delete'/'Display'/'Exit' : Exit
```

55. Write a script that accepts a number and prints its multiplication table from 1 to 10.

```
#!/bin/bash

num=$1

for((i=1;i<=10;i++))
do
echo "$(($num*$i))"
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 55.sh 4
4
8
12
16
20
24
28
32
36
40
```

56. Write a script that accepts a string and determines whether it is a palindrome.

```
#!/bin/bash

str="$1"
rev_str=$(echo "$str" | rev)

if [ "$str" = "$rev_str" ]; then
echo "$str is palindrome"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 56.sh abcba
abcba is palindrome
```

57. Write a script that counts the number of vowels in a string using a loop.

```
#!/bin/bash

str="$1"

len=${#str}
count=0

for ((i=0;i<len;i++))
do
char="${str:$i:1}"
case "$char" in
[aAeEiIoOuU]) ((count++));;
esac
done

echo "the number of vowels is $count"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 57.sh abcba
the number of vowels is 2
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

58. Write a script that accepts a filename and displays its first 5 and last 5 lines.

```
#!/bin/bash

file="$1"

if [ ! -f "$file" ]; then
echo "Invalid file"
exit 1
fi

echo "first five lines are : "
head -n 5 $file
echo "last five lines are : "
tail -n 5 $file
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 58.sh 57.sh
first five lines are :
#!/bin/bash

str="$1"

len=${#str}
last five lines are :
[aAeEiIoOuU]) ((count++));;
esac
done

echo "the number of vowels is $count"
```

59. Write a script that searches for a given word in a file and prints whether it was found.

```
#!/bin/bash

file="$1"
word="$2"

if [ ! -f "$file" ]; then
echo "Invalid file"
exit 1
fi

if grep -iq "$word" "$file"; then
echo "found"
else
echo "not found"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 59.sh 57.sh file
not found
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

60. Write a script that accepts a filename and counts the number of words, lines, and characters in it.

```
#!/bin/bash

file="$1"

words=$(wc -w < "$file")
lines=$(wc -l < "$file")
chars=$(wc -m < "$file")

echo "number of words : $words"
echo "number of lines : $lines"
echo "number of characters : $chars"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 60.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 60.sh 59.sh
number of words : 29
number of lines : 16
number of characters : 164
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

61. Write a script that displays the largest and smallest number from a list of numbers entered by the user.

```
#!/bin/bash

read -p "enter the list of numbers : " -a nums

len=${#nums[@]}
min=${nums[0]}
max=${nums[0]}
for ((i=0;i<len;i++))
do
if [ $max -lt ${nums[i]} ]; then
max=${nums[i]}
fi
if [ $min -gt ${nums[i]} ]; then
min=${nums[i]}
fi
done

echo "Max : $max and Min : $min"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 61.sh
enter the list of numbers : 1 2 3 4 56
Max : 56 and Min : 1
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

62. Write a script that accepts a number and determines whether it is prime.

```
#!/bin/bash

num=$1

if [ $num -lt 2 ]; then
echo "not a prime"
fi

for((i=2;i*i<=num;i++))
do
if [ $(num%i) -eq 0 ]; then
echo "the number is not prime"
exit 1
fi
done

echo "$num is a prime number"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 62.sh 71
71 is a prime number
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

63. Write a script that prints all prime numbers between 1 and N.

```
#!/bin/bash

num=$1

for((j=1;j<=num;j++))
do
is_prime=1
for((i=2;i*i<=j;i++))
do
if [ $(j%i) -eq 0 ]; then
is_prime=0
break
fi
done
if [ $is_prime -eq 1 ]; then
echo "$j"
fi
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 63.sh 10
1
2
3
5
7
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

64. Write a script that accepts a filename and creates a backup with the extension .bak.

```
#!/bin/bash

file="$1"
Backup_file="${file}.bak"

cp "$file" "$Backup_file"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ ls test.txt test.txt.bak
test.txt  test.txt.bak
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

65. Write a script that checks whether a file has been modified today using appropriate file-test or command-line utilities.

```
#!/bin/bash

file="$1"

today=$(date +%Y-%m-%d)
file_date=$(date -r "$file" +%Y-%m-%d)

if [ "$today" = "$file_date" ]; then
echo "Yes, the $file was modified today"
else
echo "No, $file was not modified today"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ nano 65.sh
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 65.sh 64.sh
Yes, the 64.sh was modified today
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

66. Write a script that accepts a directory and deletes all .tmp files after asking the user for confirmation.

```
#!/bin/bash

dir="$1"

for file in "$dir"/*.tmp;
do
read -p "Do you want to delete $file (y/n) : " r
if [ "$r" = "y" ]; then
rm "$file"
fi
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 66.sh ~/assignmnt4
Do you want to delete /home/pgcp-esd/assignmnt4/*.tmp (y/n) : y
rm: cannot remove '/home/pgcp-esd/assignmnt4/*.tmp': No such file or directory
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

67. Write a script that renames all .txt files in the current directory by changing their extension to .bak.

```
#!/bin/bash

dir="$1"

for file in "$dir"/*.txt;
do
if [ -f "$file" ]; then
mv "$file" "${file}.bak"
fi
done
```

68. Write a script that reads a list of numbers from a file and calculates their sum using a loop.

```
#!/bin/bash

file="$1"
sum=0

while IFS= read -r line;
do
for number in $line;
do
if [ -n "$number" ]; then
sum=$((sum+number))
fi
done
done < "$file"

echo "the sum of the numbers is $sum"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 68.sh numbers.txt
the sum of the numbers is 6389
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

69. Write a script that reads a file containing student marks and prints only students whose marks are greater than 75.

```
#!/bin/bash

file="$1"

while read -r name mark;
do
if [[ "$mark" =~ ^[0-9]+$ ]]; then
if [ "$mark" -gt 75 ]; then
echo "$name - $mark"
fi
fi
done < "$file"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 69.sh numbers.txt
b - 98
d - 76
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

70. Write a script that accepts a username and checks whether that user exists in /etc/passwd using grep.

```
#!/bin/bash

user="$1"

if grep -q "^$user" /etc/passwd; then
echo "user exists"
else
echo "user does not exists"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 70.sh dev1
user exists
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

71. Write a script that displays the current date, current user, current working directory, and number of running processes.

```
#!/bin/bash

echo "date : $(date)"
echo "user : $(whoami)"
echo "current directoy $(pwd)"
count=$(( $(ps -e | wc -l) - 1 ))
echo "number of running processes: $count"
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 71.sh
date : Monday 14 September 2026 11:37:47 PM IST
user : pgcp-esd
current directoy /home/pgcp-esd/assignment4
number of running processes: 361
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

72. Write a script that repeatedly displays a simple menu until the user selects Exit.

```
#!/bin/bash

while true;
do
read -p "enter 'Exit' to exit " input
case "$input" in
Exit) exit 1;;
*) echo "not exit"
esac
done
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 72.sh
enter 'Exit' to exit da
not exit
enter 'Exit' to exit afd
not exit
enter 'Exit' to exit Exit
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

73. Write a script that accepts a filename and reports whether it is empty or contains data.

```
#!/bin/bash

file="$1"

size=$(wc -c < "$file")

if [ "$size" -gt 0 ]; then
echo "the file is not empty"
else
echo "the file is empty"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 73.sh 72.sh
the file is not empty
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```

74. Write a shell script that monitors the disk usage of / every 10 seconds and prints a warning when usage exceeds 80%.

```
#!/bin/bash

threshold=80

while true;
do
usage=$(df -P / | awk 'NR==2 {print $5}' | tr -d '%')

if [ "$usage" -gt "$threshold" ]; then
echo "disk usage exceeded 80%"
fi
sleep 10
done
```

75. Write a shell script that asks the user for a command, executes it, and prints whether the command completed successfully or returned an error.

```
#!/bin/bash

read -p "enter a command : " comm

eval "$comm"
exit_code=$?

if [ "$exit_code" -eq 0 ]
then
echo "command executed successfully"
else
echo "command failed"
fi
```

```
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$ /bin/bash 75.sh
enter a command : pwd
/home/pgcp-esd/assignment4
command executed successfully
pgcp-esd@pgcpesd-Precision-3640-Tower:~/assignment4$
```